

Run It. Then Measure It.

Open lab: local models, variance, and the Foreman/Worker pair

Computer Science II • Dr. Jeremy P. Evert

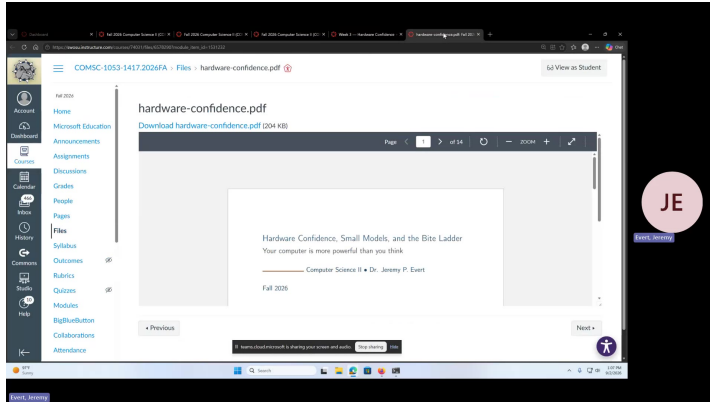
September 2, 2026

Today's destination

TODAY'S DESTINATION

- Actually get a local model talking back to you, not just installed.
- Split Foreman/Worker roles and use them on purpose.
- Turn one World Bible scenario into a real prompt.
- Measure a model, not just watch it: variance and load time.

Pick up where Monday left off



Monday's recording and slides stay linked in this module – use them alongside today's lab.

Foreman, Worker, Navigator

- Worker drives the keyboard. Foreman narrates intent before acting.
- Foreman's real job: catch the moment the Worker notices something interesting and *doesn't* stop to ask about it.
- That moment is exactly what gets pasted into Copilot or ChatGPT.
- Bonus points for the Foreman: get artifacts into GitHub, and keep lab notes.

Four worlds, four seeded inconsistencies

Step 2 — Answer a call

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Here are four worlds to choose from:

Frontier Settlement

A growing settlement whose informal record-keeping can no longer keep up. The mill's grain count has started coming up short.

Investigation Bureau

A records bureau where evidence, reports, cases, people, and timelines must agree. One report contains a timestamp that doesn't match.

Starship Log

A research vessel operating far from support. The crew must trust its operational systems. One sensor has been behaving strangely.

Small Business

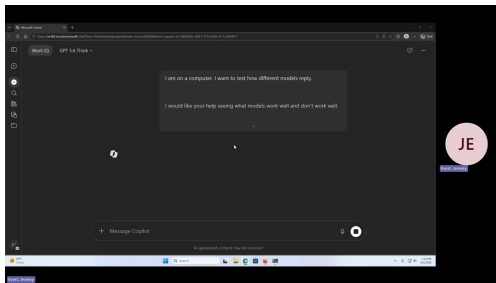
A successful hardware store that has outgrown handwritten notes and information stored in people's heads. Inventory has come up short twice.

Then:

Choose the world you are most curious to return to for the rest of the semester. There is no harder or easier choice.

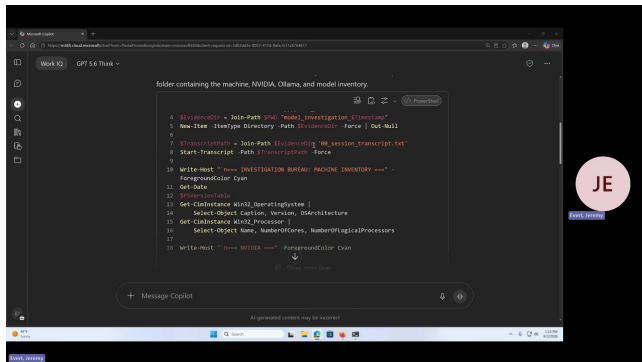
Frontier Settlement, Investigation Bureau, Starship Log, Small Business – each one ships with a mismatch to chase.

Ask for a script – then read it



- Plain-language ask: “I am on a computer, I want to test how different models reply.”
- Copilot handed back a PowerShell scaffold.
- The habit that matters: read every line before you run it.

What the scaffold actually did



folder containing the machine, NVIDIA, Ollama, and model inventory.

```
4 $EvidenceDir = Join-Path $PWD "model_investigation_$timestamp"
5 New-Item -ItemType Directory -Path $EvidenceDir -Force | Out-Null
6
7 $TranscriptPath = Join-Path $EvidenceDir "00_session_transcript.txt"
8 Start-Transcript -Path $TranscriptPath -Force
9
10 Write-Host "==== INVESTIGATION BUREAU: MACHINE INVENTORY ===" -
    ForegroundColor Cyan
11
12 Get-Date
13 $VersionTable
14 Get-CimInstance Win32_OperatingSystem |
    Select-Object Caption, Version, OSArchitecture
15 Get-CimInstance Win32_Processor |
    Select-Object Name, NumberOfCores, NumberOfLogicalProcessors
16
17 Write-Host "==== NVIDIA ===" ForegroundColor Cyan
18
```

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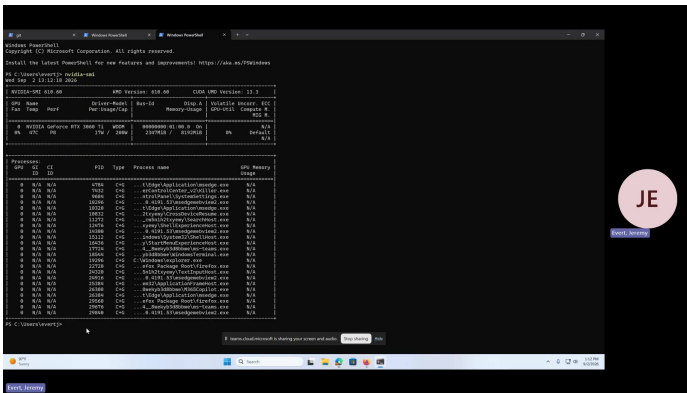
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Message Copilot

AI generated content may be incorrect

Start-Transcript to log the session, Get-CimInstance for machine info, an nvidia-smi block for GPU state.

Read `nvidia-smi` before you blame the model



VRAM already in use at idle is real – check headroom before deciding a model “doesn’t fit.”

Variance is measurable, not a vibe

- Ask the same question five times – describe how the answers differ.
- Ask the same question three times – time each response.
- **Cold load:** weights are being pulled into VRAM for the first time.
- **Hot load:** already resident, answers faster.
- This is where the semester's later Computer Architecture / memory conversation picks back up.

Before next class

RECAP

- A model download is network/disk-bound – pipeline a second pull in a second window instead of waiting.
- Foreman bonus: durable GitHub artifacts and real lab notes, not just a working demo.
- “It runs” is still not the same claim as “it runs usefully, and I can describe how much it wiggles.”